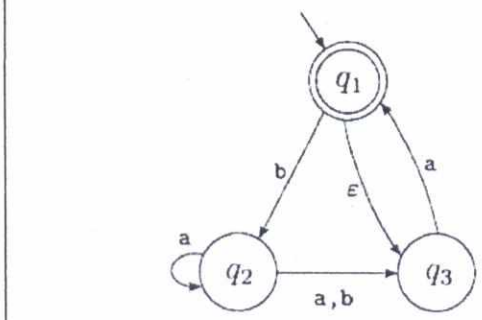
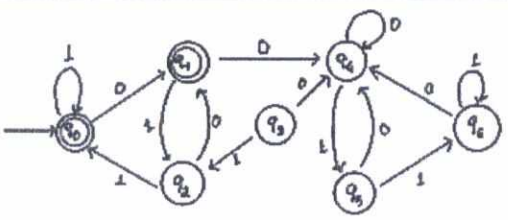
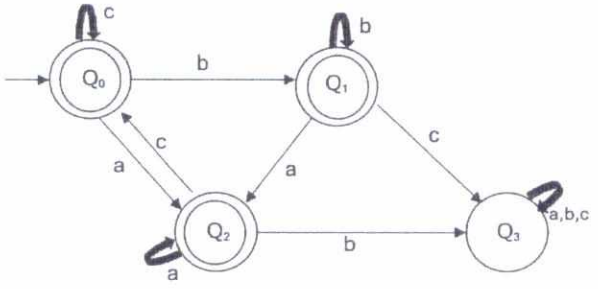
Roll Number: _____ Name_____ Group_____

THAPAR INSTITUTE OF ENGINEERING &TECHNOLOGY, PATIALA
Department of Computer Science & Engineering
Theory of Computation (UCS701) – Mid Semester Examination
Date: 18/03/2026 & Time: 08:00AM MM: 30 & MT: 120 Min

List of faculties: Dr. Chinmaya Panigrahy (C2), Dr. Amrita Kaur (C1), Dr. Aditi Sharma (C6, P1), Dr. Vibha Jain (C3, Q2), Dr. Nidhi Chakravarty (C7, Q1), and Dr. Nitin Arora (C4, C5)

Attempt all the questions. Attempt/Answer all sub-parts like (a), (b), (c) for each question in one place. Do mention Page No. of your attempt on the front page. Question paper is of two pages. Please turn over this page to see the questions on the next page.

Ques. No		Questions	Mark	CO	BL
Q1.	(a)	Apply Thompson's construction to construct an NFA with epsilon moves for the regular expression, ((a + b)*ab + ba)*.	2	CO1	L3
	(b)	Design a minimal Mealy machine for the following. "For input from (0+1)*, if the input ends in 101, machine outputs A; if input ends in 110, machine outputs B, otherwise machine outputs C." Convert the resultant Mealy machine to an equivalent Moore machine.	3	CO1	L3
	(c)	List all the minimal length strings not in the language represented by the regular expression, a*(baa*)*b*.	1	CO1	L2
	(d)	Convert the given NFA with epsilon moves to an equivalent NFA. 	1.5	CO1	L3
	(e)	Construct an NFA for the regular expression, (a+b)*(aa+bb) (a+b)* by showing the intermediate steps and convert the NFA into a resultant DFA.	2.5	CO1	L4
Q2.	(a)	Design a simple computer having two switches, which lights a bulb when both the switches are in ON mode. Initially, both the switches are in OFF mode.	1.5	CO1	L3
	(b)	Design a minimal DFA to accept all the binary strings that begin and end with 101. Write the corresponding regular expression. Write down the tuple representation of the resultant DFA.	3	CO1	L3
	(c)	Prove or disprove the following statement. "Regular languages are closed under concatenation operation."	1	CO1	L3
	(d)	Minimize the given DFA. 	2.5	CO1	L3

	(e)	In a factory, a conveyor belt connects two machines M_1 and M_2. Machine M_1 places items onto the conveyor and machine M_2 removes items from the conveyor. The conveyor can hold at most 3 items at a time. A computer records an input symbol: • **push** whenever an item is placed onto the conveyor (by M_1) • **pop** whenever an item is removed from the conveyor (by M_2) Assume that at the beginning and end, the conveyor is empty. Design a **DFA** over the alphabet $\Sigma = \{\text{push}, \text{pop}\}$ that accepts exactly those sequences where: 1. The number of items on the conveyor never becomes negative. 2. The number of items on the conveyor never exceeds 3. 3. The conveyor ends empty.	**2**	**CO1**	**L3**
Q3.	**(a)**	State and Prove the Arden's theorem.	**3**	**CO1**	**L4**
	(b)	Find the regular expression corresponding to the given finite automaton using Arden's theorem.  	**2.5**	**CO1**	**L3**
	(c)	Write a regular expression to represent all the binary strings (i) That does not contain two consecutive 0's and begin with 0. (ii) That does not contain two consecutive 0's and end with 0.	**1**	**CO1**	**L3**
	(d)	Consider the language, L= $\{n_a(w) < n_b(w)$: w is a string in Σ^*, whereas $n_a(w)$ and $n_b(w)$ represent the number of a's and b's present in w, respectively$\}$. If L is regular, justify your answer by designing a DFA. Otherwise, use the Pumping lemma for regular languages to prove that L is non-regular. Here, $\Sigma = \{a,b\}$.	**2.5**	**CO1**	**L4**
	(e)	Devansh was asked to create a set, S, by taking some of the strings from the language represented by the regular expression, (a+b)*. Is it possible to say that S is always regular? Justify your answer.	**1**	**CO1**	**L4**